



AWS Certified AI Practitioner

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AI and ML fundamentals

AI: Machines that performs jobs that mimic human behaviour.

Machine Learning: Machines that get better at a task without being explicitly programmed.

Deep Learning: Machines that use artificial neural networks inspired by human brain to solve complex tasks.

GenAI: Specialized subset of AI that generate content e.g. text, image, video

AI performs tasks that often require human intelligence. Its goal is to interpret, analyze and respond to human action to simulate human intelligence in machines.

GenAI focuses on creating new content or data that is novel and realistic. It can interpret and analyze data but also it can generate new data itself. GenAI has multiple modalities like text, vision, audio (think of it as human senses). LLM is one modality of GenAI that generates natural language text.

NLP is a ML technique that understands the context of a corpus(a body of related text). Analyze and interpret text within documents, spoken words and language translation and perform appropriate actions based on the interpretation.

Regression is a process of finding a function to correlate a labeled dataset to a continuous variable/number. The distance of the vector from regression line is

known as error. Classification is a process of finding a function to divide the labeled dataset into classes/categories. Clustering is the process of grouping unlabeled data based on similarities and differences.

Classification algorithms: logistic regression, knearest neighbours, support vector machines (SVM), Naive bayes, decision tree, random forest

Regression algorithms: Linear, multiple linear, polynomial, SVM, decision tree, random forest

Clustering is the process of grouping. unlabeled data based on some similarities and differences, algorithms are k-means clustering, DBSCAN. Association is the process of finding relationships between data through association, it is used in recommendation systems, algorithms are apriori, euclat, FP-growth.

Dimensionality reduction reduce the numbers of features while retaining data integrity, mostly used for data analysis and visualisation, algorithms are Principle Component Analysis(PCA), Single Value Decomposition, t-SNE. All of these are part of unsupervised learning.

Type of Machine Learning

- Supervised: Have a labeled dataset and learned a mapping between the input and the output variables, task driven, makes a prediction
- Unsupervised: Using a model to describe a relationship between an unlabeled data, data driven, recognize a structure or pattern, you are trying to make sense of data
- Reinforcement: An agent operate in an environment and learns to operate and improve , based on some feedback
- Semi Supervised: Mix of labeled and unlabeled data, mostly have unlabeled data and few labeled data
- Self-supervised: It can label its own data in order to apply supervised learning algorithms
- Multi-instance: Individual examples are unlabeled but groups/bags of them are labeled
- Inductive: Using specific evidence to determine outcomes

- Deductive: Using general rules to determine outcomes
- Transductive: Used in statistical theory to predict specific examples given from a specific domain
- Multitask: Fitting a model on one dataset that addresses multiple related problems
- Active: The model is able to query a human operator during the learning process in order to resolve ambiguity
- Online: Using the available data and updating the model before a prediction is required
- Transfer: A model is trained on one task and then the model is used as the starting point/base model for another task

Divisions of Machine Learning

- Classical Machine Learning: Supervised machine learning where we have labeled or unlabeled data and use statistical techniques
- Reinforcement learning: An agent improves itself by feedback, used when there is no data
- Ensemble learning: Used when quality of data is a problem e.g. bagging, boosting, stacking
- Neural Network: When data is complicated and hard to interpret, has many features e.g. CNN, RNN, GAN, multilayer perceptron, auto encoders

Neural Network/Deep Learning

Neural network having more than 3 or more hidden layers is considered deep learning.

In Feed forward neural network nodes don't have a cycle so data always moves forward.

Backward propagation is the process of moving backward through neural network, adjusting weights to improve the outcome of algorithm on next iteration.

Loss function compare the actual output to the algorithm's output to determine the error rate.

Activation function: Algorithm applied to one of the hidden nodes that affects connected outputs. It decide if the output of it will proceed to the next layer. A node can be active if the output moves to next layer. Types of activation function: Linear(cannot backpropagate), binary(if greater than threshold then pass, returns 0 or 1, sigmoid(range between 0 and 1, have vanishing gradient problem), tanh(have range b/w -1 and 1, has vanishing gradient problem), ReLu(suppress negative value), Leaky ReLu(negative slope is not linear so no dying ReLu problem), softmax(converts raw vector into probability distribution), Exponential Linear Unit(ELU) slops/dies towards -1 so it avoids dying ReLu problem to some extent

When layer increases(amount of nodes increases) the network is dense and when decreases then it is sparse

ML model

- An algorithm consists of mathematical or computational steps to perform a specific task
- A model is an informative representation of an object or a person. A model can be concrete(person, object) or abstract(computer algorithm, mathematics model)
- ML model is a function that gets an input data and performs prediction. A model is trained on training data and then deployed and then unlabeled data is provided to perform prediction based on inference
- Feature is a characteristic extracted from unstructured data ingested by the ML model to perform predictions. ML models accept numerical data so we convert our data into machine readable format by **encoding**. Feature engineering is method of extracting features from data
- **Model Parameter** represents an internal state of model which is not manually set, their values are estimated and learned during training of mode. They are used to make predictions. **Hyperparameter** is external to the model which are manually set before training of model and used during training of model e.g. learning rate, epoch, batch size

Responsible AI

<https://aws.amazon.com/ai/responsible-ai/>

Data Labelling

Process of manually adding meaningful one or more labels/context to raw data so that model can learn from. Labels can be single word(classification) or bounding boxes(object detection). **Ground truth** is a properly labeled dataset that you as a standard to train and test a model. Accuracy of model depends on the accuracy of ground truth

Data Mining

Process of extracting patterns and knowledge from large amount of data instead of extracting the data. There are standard steps for performing data mining:

1. Business needs: what do the business needs?
2. Data understanding: what data do we currently have and what do we need?
3. Data preparations: how do we structure the data for modeling
4. Modeling: what modeling techniques should we apply
5. Evaluations: how do we evaluate the data or model that it meets the business objectives?
6. Deployment: how to people access data?

Data mining techniques like classification, clustering, association methods, regression are used to identify patterns and derive insights from large amounts of data

Data wrangling/modeling

Process of transforming or mapping data from one format to another with intent of making it more appropriate and usable for downstream services is called data wrangling. Simply it means preprocessing/cleaning your data. It has below four steps:

1. discovery: understand your data along with its domain specific details

2. structuring: organise your data so that it is easier to process it
3. cleaning: removing null values, duplicates, outliers, standardize format
4. enriching: appending additional information and context to data obtained from additional sources
5. validating: authenticate reliability, quality and safety of data
6. publishing: uploading your data to datastore so other services can use it

A data model is an abstract organization of data elements and how they relate to one another and to properties of real world e.g. people, cars etc. A data model can be conceptual(people, animals, relationships), logical(tables, foreign keys), physical(partition, OS pages etc). Data modeling is the method is analyzing data requirements needed to support the business confined within the scope of organisation

Data analytics is concerned with analyzing, studying and transforming data to extract and study useful information. Ingest → clean → dimensionality reduction → data analysis → visualization

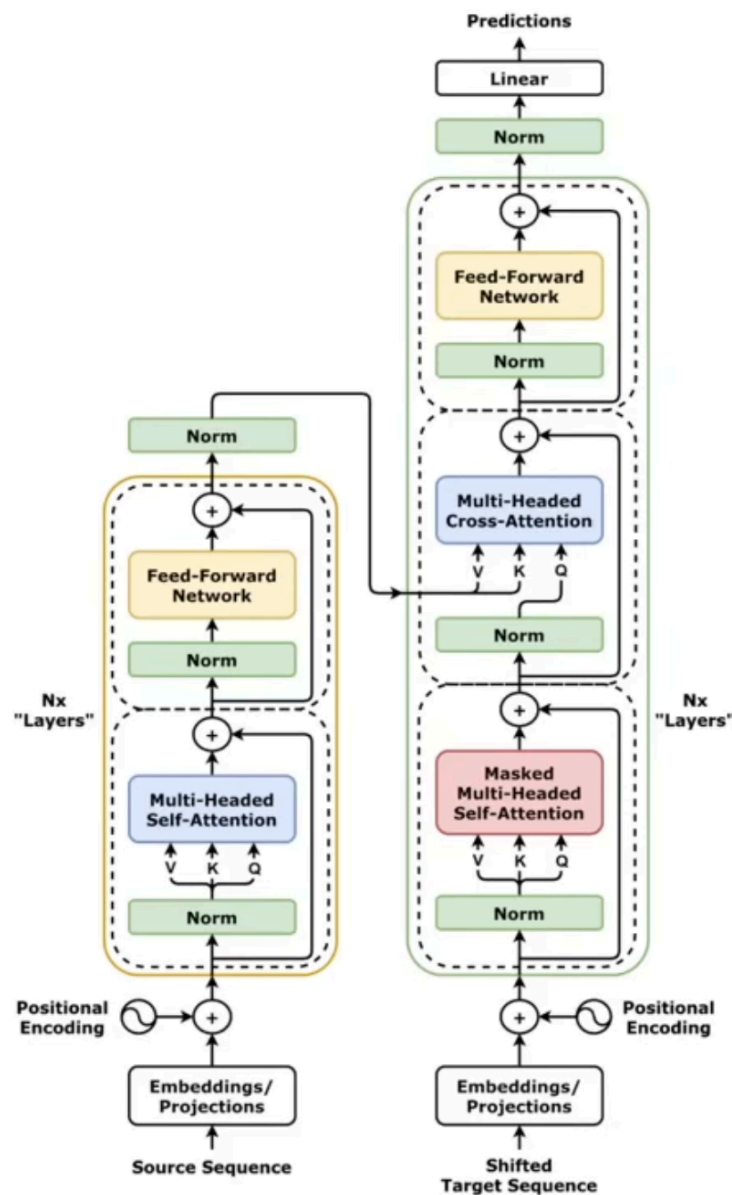
Data Type

- Qualitative → Categorical → Discrete → binary(0,1/Yes,No), nominal(without order e.g. colors blue, green white), ordinal(ordered e.g. poor, middle class, rich)
- Quantitative → numerical(countable → continuous(not countable as multiple measurement values) → interval, ratio

AI/GenAI

- AI is used to perform tasks that requires human intelligence. It simulate(mimics, similar behaviour) but not emulate human intelligence. It can analyze, interpret and make decisions based on existing data
- Generative AI is usually used to create new data or content that is novel and realistic based on the existing data. It can also analyze and interpret data but generally used for data generation. It has multiple modalities e.g. text, vision, audio etc. Popular techniques involves, GANs, VAEs, Transformers/GPTs

- Foundational models are general purpose models that are trained on large amount of data and are further finetuned to perform specific tasks. LLMs are foundational models that implements transformer architecture
- Transformer architecture was developed by researchers at google that is used at NLP using multi head attention and positional encoding



- Tokenization is the process of breaking down input data(text) into smaller pieces so that it can be fed into the LLM as part of its internal vocabulary. Decoder feeds its output again as input so that it can produce the next token,

as the number of time decoder feeds the output, the memory to store and the compute requires to perform operation on those additional tokens increases

- Embeddings are vectors of data used by ML models to draw relationships between data. Separate ML models are trained to create embeddings. They project data into the vector space with a magnitude/length and direction
- Positional encoding is the process to preserve order of words/tokens since processing data sequentially would lose the order of words to understand the input. It adds a positional vector to each word/token to keep track of its order
- Attention figures out how important a word/token is in the sequence with respect to other tokens and assigns a weight to each one. Self attention computes weights wrt tokens within the same sequence whereas cross attention computes weights between the input(encoder) and output(decoder) sequences tokens. Multihead attention combines multiple attention in parallel and is used in transformers
- Finetuning is the process of retraining a pretrained model on a smaller dataset to make it better on some specific tasks. Supervised finetuning(SFT) is when the retraining data is labeled. Instruction finetuning is when the dataset has both good and bad instructions or do's or don't's. Parameter efficient finetuning(PEFT) is when only a limited number of parameters are updated. LORA is an example of PEFT. Last layer finetuning is between full finetuning and PEFT that updates and trains only the last layer of the model

Amazon Bedrock

- Amazon Bedrock provides LLM models as a service with other bunch of functionalities:
 - Model Catalog
 - Finetuning of custom local LLMs
 - Prompt Management
 - Knowledge base - RAG
 - Agents and agentic workflows

- Guardrails
- Evaluation assessments
- Bedrock has two types of deployment methods, Not all models offers both deployment method:
 - On-demand(serverless): You are paying based on input and output tokens. You can track token usage and cloudwatch logs and metrics
 - Provisioned Throughput: You are paying based on model units(MUs) which provides a dedicated level of throughput for specific models. Its cost cannot be zero since the server is always running even though you not using any tokens
- Batch Inference: Non-realtime bulk request processing for selected small models. Cross-Region Inference: Route request to multiple AWS regions during peak utilization

Knowledge Base

- It allows you to setup a RAG workflow to a vectorstore. Vector store doesn't necessarily means a graph database, it can be any relational, document, key-value database as well
- data sources → chunking → parsing → embedding → vectorstore
- Data sources can be S3, web crawler, salesforce, sharepoint, google drive. There are multiple chunking options such as fixed, hierarchical and semantic chunking and you can also use lambda functions. Parsing is used to read or input text from multiple formats like txt, pdf, docx, html in a standard format for storing. Vectorstore options are amazon aurora(using pg_vector), mongodb, pinecone etc